End-Term Sample Question Paper - ANSWER KEY

Course Code: CSE563 Course Title: APPLIED MATHEMATICS

Maximum Marks: 60 Duration: 2 Hours Set C - all six units, theory and numerical

Academic Task Type: End-Term Sample Date: _____ Registration Number: _____

Instructions: answer all questions. Part A carries 2 marks each, Part B 5 marks each and Part C 10 marks each; in Part C the marks of each sub-part are shown in brackets. Show all working for numerical parts — an unsupported final answer earns no marks. The unit, course outcome and Bloom level are printed against every question.

For the instructor. Every numerical result in this key was recomputed and verified before printing; the marks in brackets give the part-wise marking scheme.

Part A - Very Short Answer Questions (5 x 2 = 10 Marks)

Q1. State De Morgan's laws for sets, and write the complement of A ∪ B in words. **(2 Marks)**

[Unit I · CO1 · L1 - Remember · Theory]

$(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$. (1.5 marks)

In words: the elements outside A ∪ B are exactly those lying **outside A and outside B**. (0.5 marks)

Q2. Calculate the eigenvalues of A = [[3, 1], [1, 3]], and verify them with the trace and the determinant. **(2 Marks)**

[Unit II · CO2 · L2 - Understand · Numerical]

tr A = 6, det A = 9 − 1 = 8, so $\lambda^2 - 6\lambda + 8 = 0$, giving **$\lambda = 4$ and $\lambda = 2$**. (1.5 marks)

Check: 4 + 2 = 6 = tr A and 4 × 2 = 8 = det A. (0.5 marks)

Q3. A server receives requests following a Poisson distribution with mean $\lambda = 2$ per minute. Compute P(X = 1), correct to four decimal places. **(2 Marks)**

[Unit III · CO3 · L2 - Understand · Numerical]

$P(X = k) = e^{-A}\lambda^k/k!$, so $P(X = 1) = e^{-2}(2)/1!$. (1 mark)

$= 0.135335 \times 2 =$ **0.2707**. (1 mark)

Q4. State the condition for a connected graph to have an Eulerian circuit, and the condition for it to have an Eulerian path but no circuit. **(2 Marks)**

[Unit IV · CO4 · L1 - Remember · Theory]

An Eulerian **circuit** exists iff **every vertex has even degree**. (1 mark)

An Eulerian **path** but no circuit exists iff **exactly two vertices have odd degree**; the path must start at one and end at the other. (1 mark)

Q5. State what it means for a problem to be in P and in NP, and state what the Cook–Levin theorem established. **(2 Marks)**

[Unit V · CO5 · L2 - Understand · Theory]

P: solvable in polynomial time by a deterministic machine. **NP:** a proposed certificate can be **verified** in polynomial time. (1 mark)

Cook–Levin established that **SAT is NP-complete** — the first such problem, and the starting point every later NP-completeness proof reduces from. (1 mark)

Part B - Short Answer Questions (4 x 5 = 20 Marks)

Q6. Apply the 0/1 knapsack method to items of weights {3, 4, 5} and values {₹30, ₹50, ₹60} with capacity W = 8. List the feasible subsets, state the optimal value and the items chosen. **(5 Marks)**

[Unit VI · CO6 · L3 - Apply · Numerical]

Feasible subsets (weight ≤ 8): {1} 30; {2} 50; {3} 60; {1,2} weight 7, value 80; {1,3} weight 8, value **90**. (3 marks)

{2,3} weighs 9 > 8, so it is infeasible. (0.5 marks)

Optimal value = **₹90**, taking **items 1 and 3** (weight 3 + 5 = 8, exactly filling the knapsack). (1.5 marks)

Q7. In a survey of 120 students, 60 use Windows, 45 use Linux and 40 use macOS; 25 use Windows and Linux, 20 use Windows and macOS, 15 use Linux and macOS, and 10 use all three. Using the inclusion–exclusion principle, find how many use at least one of the three, and how many use none. **(5 Marks)**

[Unit I · CO1 · L3 - Apply · Numerical]

$|W \cup L \cup M| = 60 + 45 + 40 - 25 - 20 - 15 + 10$. (2 marks)

$= 145 - 60 + 10 =$ **95** students use at least one. (2 marks)

None: $120 - 95 =$ **25** students. (1 mark)

Q8. Distinguish a confidence interval from a hypothesis test, define Type I and Type II errors, and state the relationship between a 95 per cent interval and a two-tailed test at $\alpha = 0.05$. **(5 Marks)**

[Unit III · CO3 · L2 - Understand · Theory]

A **confidence interval** gives a range of plausible values for the parameter; a **hypothesis test** returns a decision about one specific claimed value. (2 marks)

Type I error: rejecting a true null hypothesis (a false positive, probability α). **Type II** error: failing to reject a false null (a false negative, probability β). (2 marks)

They agree exactly: a two-tailed test at $\alpha = 0.05$ rejects H_0 **precisely when** the null value falls outside the 95 per cent confidence interval. (1 mark)

Q9. Describe three methods for solving a recurrence relation — iteration, the recursion tree, and the characteristic equation — and state which kind of recurrence each suits. **(5 Marks)**

[Unit V · CO5 · L4 - Analyse · Theory]

Iteration (substitution): unroll the recurrence until the base case and sum the terms. Suits simple decreasing recurrences such as $T(n) = T(n-1) + f(n)$. (2 marks)

Recursion tree: draw the work at each level and sum level by level. Suits divide-and-conquer forms $T(n) = aT(n/b) + f(n)$, and motivates the Master theorem. (1.5 marks)

Characteristic equation: for a linear recurrence with constant coefficients, solve $r^2 - c_1 r - c_2 = 0$ and fit the constants to the initial conditions. Suits forms such as $a_n = c_1 a_{n-1} + c_2 a_{n-2}$. (1.5 marks)

Part C - Long Answer Questions (3 x 10 = 30 Marks)

Q10. Image transformation in homogeneous coordinates. **(10 Marks)**

[Unit II · CO2 · L4 - Analyse · Theory + Numerical]

(a) Explain why a translation cannot be written as a plain 2×2 matrix and how homogeneous coordinates solve this, and state why the order of composition matters. *[5]*

(b) A point is first translated by (3, 2) and then scaled by 2 about the origin. Write T and S as 3×3 matrices, compute $M = S \cdot T$, and apply M to the point (1, 4), showing each stage. *[5]*

(a) A 2×2 matrix acts as a **linear** map: it satisfies $M0 = 0$, so it can never move the origin and cannot add a constant offset. Translation is **affine**, not linear. (2 marks)

Appending a third coordinate 1 and using $[[1,0,t_x],[0,1,t_y],[0,0,1]]$ turns the offset into ordinary matrix multiplication, so rotations, scalings and translations compose into a **single** 3×3 matrix applied in one pass. (2 marks)

Matrix multiplication is **not commutative**: scaling after translating multiplies the offset too, so $S \cdot T \neq T \cdot S$. (1 mark)

(b) $T = [[1,0,3],[0,1,2],[0,0,1]]$, $S = [[2,0,0],[0,2,0],[0,0,1]]$. (2 marks)

$M = S \cdot T =$ **[[2, 0, 6], [0, 2, 4], [0, 0, 1]]**, i.e. $(x, y) \mapsto (2x + 6, 2y + 4)$. (1.5 marks)

Stages for (1, 4): translate → (4, 6); scale → **(8, 12)**. Directly: $(2(1) + 6, 2(4) + 4) = (8, 12)$. Correct. (1.5 marks)

Q11. Graph colouring and timetabling. **(10 Marks)**

[Unit IV · CO4 · L4 - Analyse · Theory + Numerical]

(a) Define the chromatic number, state the bound involving the size of the largest clique, and describe the Welsh–Powell heuristic. *[5]*

(b) Five papers A to E must be scheduled; the pairs sharing students are A–B, A–C, B–C, B–D, C–E and D–E. List the degrees, apply Welsh–Powell, and give the timetable with the number of slots. *[5]*

(a) The **chromatic number** $\chi(G)$ is the least number of colours needed so that adjacent vertices differ. Since every vertex of a clique must take a distinct colour, $\boldsymbol{\chi(G) \geq \omega(G)}$, the size of the largest clique. (3 marks)

Welsh–Powell: sort the vertices by decreasing degree; take the first uncoloured vertex, give it a new colour, and sweep the rest of the list giving that same colour to every vertex not adjacent to any already coloured with it; repeat. It is a heuristic — it uses few colours but is not guaranteed optimal. (2 marks)

(b) Degrees: A = 2, B = 3, C = 3, D = 2, E = 2 (sum 12 = 2 × 6 edges, check). (1.5 marks)

Order by decreasing degree: B(3), C(3), A(2), D(2), E(2). (0.5 marks)

Colour 1: B; A, C and D all conflict with B, but **E** does not → {B, E}. Colour 2: C; A conflicts with C, **D** does not → {C, D}. Colour 3: **A**. (2 marks)

3 slots: Slot 1 {B, E}, Slot 2 {C, D}, Slot 3 {A}. Three is optimal here because A, B, C form a triangle, forcing $\chi \geq 3$. (1 mark)

Q12. Divide and conquer, and solving its recurrences. **(10 Marks)**

[Unit VI · CO6 · L5 - Evaluate · Theory + Numerical]

(a) State the divide-and-conquer template, explain how Karatsuba and Strassen each reduce the number of sub-problems, and state why that lowers the exponent. *[5]*

(b) Solve: (i) $T(n) = 2T(n/2) + n^3$; (ii) $T(n) = 7T(n/2) + n^2$; (iii) $T(n) = 2T(n-1) + 3$ with $T(0) = 1$ — give the closed form and $T(6)$. *[5]*

(a) Divide the instance into sub-problems, conquer them recursively, and combine the results — giving $T(n) = aT(n/b) + f(n)$. (2 marks)

Karatsuba multiplies two n-digit numbers with **3** half-size products instead of 4, giving $\Theta(n^{\log_2 3}) \approx \Theta(n^{1.585})$. **Strassen** multiplies 2×2 blocks with **7** products instead of 8, giving $\Theta(n^{\log_2 7}) \approx \Theta(n^{2.807})$. (2 marks)

The count of sub-problems a sits in the exponent as $\log_b a$, so reducing a from 4 to 3 (or 8 to 7) lowers the exponent directly. (1 mark)

(b)(i) $a = 2$, $b = 2$, $\log_2 2 = 1 < 3$, so $f(n) = n^3$ dominates — **Case 3**: $\Theta(n^3)$. (1.5 marks)

(ii) $a = 7$, $b = 2$, $\log_2 7 \approx 2.807 > 2$, so the sub-problems dominate — **Case 1**: $\Theta(n^{\log_2 7}) \approx \Theta(n^{2.807})$. (1.5 marks)

(iii) Unrolling: $T(1) = 5$, $T(2) = 13$, $T(3) = 29$, $T(4) = 61$. Closed form $\mathbf{T(n) = 4 \cdot 2^n - 3}$, so $T(6) = 4(64) - 3 =$ **253**. (2 marks)

----- End of Answer Key -----